

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
```

```
df = pd.read_csv("heart_ecg.csv")
print(df.shape)
print(df.head())
```

```
(58726, 188)
  signal_1  signal_2  signal_3  signal_4  signal_5  signal_6  signal_7  \
0        1.0  0.698745  0.012552  0.012552  0.150628  0.133891  0.071130
1        1.0  0.350290  0.024469  0.065035  0.141017  0.225370  0.250483
2        1.0  0.945055  0.611722  0.260073  0.150183  0.124542  0.084249
3        1.0  0.897275  0.232704  0.023061  0.079665  0.153040  0.136268
4        1.0  0.950637  0.735669  0.544586  0.530255  0.565287  0.565287

  signal_8  signal_9  signal_10  ...  signal_179  signal_180  signal_181  \
0  0.062762  0.066946  0.071130  ...          0.0          0.0          0.0
1  0.238892  0.237605  0.238249  ...          0.0          0.0          0.0
2  0.087912  0.069597  0.069597  ...          0.0          0.0          0.0
3  0.140461  0.129979  0.121593  ...          0.0          0.0          0.0
4  0.565287  0.554140  0.557325  ...          0.0          0.0          0.0

  signal_182  signal_183  signal_184  signal_185  signal_186  signal_187  \
0          0.0          0.0          0.0          0.0          0.0          0.0
1          0.0          0.0          0.0          0.0          0.0          0.0
2          0.0          0.0          0.0          0.0          0.0          0.0
3          0.0          0.0          0.0          0.0          0.0          0.0
4          0.0          0.0          0.0          0.0          0.0          0.0

  label
0      0
1      0
2      0
3      0
4      0
```

```
[5 rows x 188 columns]
```

```
X = df.iloc[:, :-1].values
y = df.iloc[:, -1].values
```

```
df.iloc[:, -1].value_counts(normalize=True)
```

	proportion
label	
0	0.5
1	0.5

dtype: float64

```
scaler = StandardScaler()
X = scaler.fit_transform(X)
```

```
X = X.reshape(X.shape[0], X.shape[1], 1)
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
```

```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Conv1D, MaxPooling1D, Flatten, Dense, Dropout
from tensorflow.keras.optimizers import Adam
from tensorflow.keras.utils import to_categorical
```

```
model = Sequential([
    Conv1D(64, kernel_size=7, activation='relu', input_shape=(187, 1)),
    BatchNormalization(),
    MaxPooling1D(2),

    Conv1D(128, kernel_size=5, activation='relu'),
    BatchNormalization(),
    MaxPooling1D(2),

    Conv1D(256, kernel_size=3, activation='relu'),
    BatchNormalization(),
    MaxPooling1D(2),

    Flatten(),
    Dense(128, activation='relu'),
    Dropout(0.5),
    Dense(1, activation='sigmoid')
])
```

```
model.compile(
    loss='binary_crossentropy',
    optimizer=Adam(learning_rate=0.005),
    metrics=['accuracy']
)
```

```

history = model.fit(
    X_train, y_train,
    epochs=20,
    batch_size=64,
    validation_data=(X_test, y_test),
    verbose=1
)

```

```

Epoch 1/20
735/735 ————— 13s 11ms/step - accuracy: 0.7166 - loss: 0.8331
Epoch 2/20
735/735 ————— 4s 5ms/step - accuracy: 0.8675 - loss: 0.3157 -
Epoch 3/20
735/735 ————— 4s 5ms/step - accuracy: 0.9154 - loss: 0.2191 -
Epoch 4/20
735/735 ————— 5s 5ms/step - accuracy: 0.9311 - loss: 0.1814 -
Epoch 5/20
735/735 ————— 4s 5ms/step - accuracy: 0.9437 - loss: 0.1477 -
Epoch 6/20
735/735 ————— 4s 6ms/step - accuracy: 0.9446 - loss: 0.1485 -
Epoch 7/20
735/735 ————— 4s 5ms/step - accuracy: 0.9554 - loss: 0.1241 -
Epoch 8/20
735/735 ————— 4s 5ms/step - accuracy: 0.9582 - loss: 0.1140 -
Epoch 9/20
735/735 ————— 4s 6ms/step - accuracy: 0.9563 - loss: 0.1190 -
Epoch 10/20
735/735 ————— 5s 5ms/step - accuracy: 0.9626 - loss: 0.1037 -
Epoch 11/20
735/735 ————— 4s 5ms/step - accuracy: 0.9667 - loss: 0.0947 -
Epoch 12/20
735/735 ————— 4s 6ms/step - accuracy: 0.9668 - loss: 0.0901 -
Epoch 13/20
735/735 ————— 4s 5ms/step - accuracy: 0.9675 - loss: 0.0894 -
Epoch 14/20
735/735 ————— 4s 5ms/step - accuracy: 0.9706 - loss: 0.0778 -
Epoch 15/20
735/735 ————— 4s 6ms/step - accuracy: 0.9725 - loss: 0.0798 -
Epoch 16/20
735/735 ————— 4s 5ms/step - accuracy: 0.9706 - loss: 0.0789 -
Epoch 17/20
735/735 ————— 4s 5ms/step - accuracy: 0.9679 - loss: 0.0889 -
Epoch 18/20
735/735 ————— 4s 5ms/step - accuracy: 0.9756 - loss: 0.0676 -
Epoch 19/20
735/735 ————— 4s 5ms/step - accuracy: 0.9732 - loss: 0.0725 -
Epoch 20/20
735/735 ————— 4s 5ms/step - accuracy: 0.9754 - loss: 0.0686 -

```

```

test_loss, test_acc = model.evaluate(X_test, y_test)
print(f"Test Accuracy: {test_acc:.4f}")

```

```

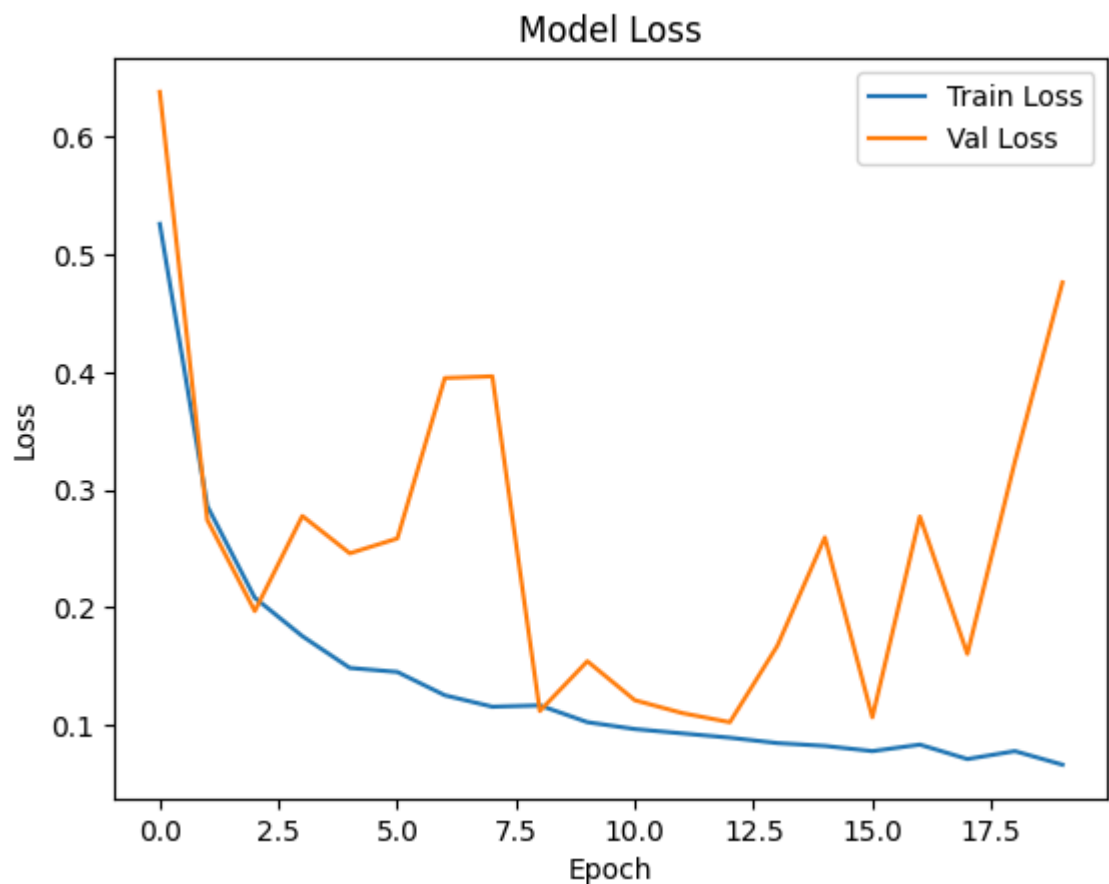
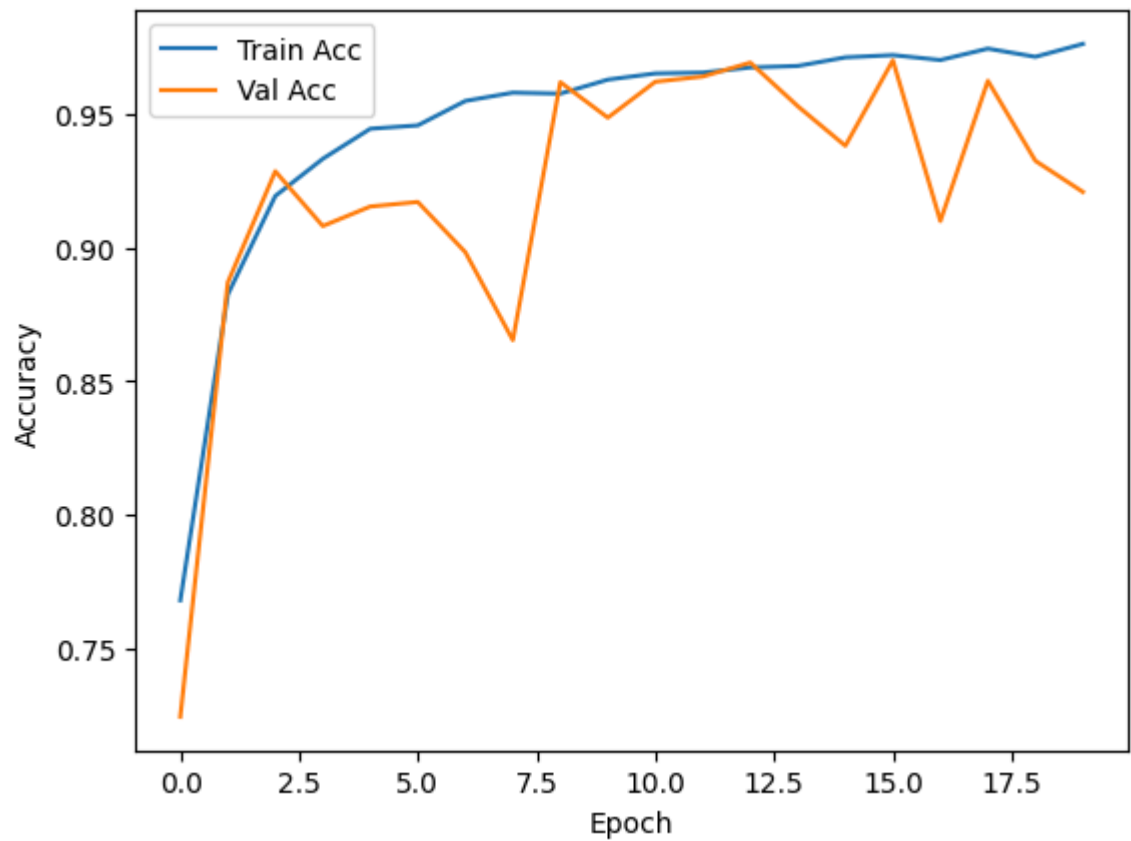
368/368 ————— 2s 3ms/step - accuracy: 0.9173 - loss: 0.4707
Test Accuracy: 0.9208

```

```
import matplotlib.pyplot as plt

plt.plot(history.history['accuracy'], label='Train Acc')
plt.plot(history.history['val_accuracy'], label='Val Acc')
plt.legend()
plt.xlabel('Epoch')
plt.ylabel('Accuracy')
plt.show()

plt.plot(history.history['loss'], label='Train Loss')
plt.plot(history.history['val_loss'], label='Val Loss')
plt.title('Model Loss')
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.legend()
plt.show()
```



```
test_loss, test_acc = model.evaluate(X_test, y_test)
print(f"✅ Test Accuracy: {test_acc:.4f}")
print(f"📄 Test Loss: {test_loss:.4f}")
```

368/368 ————— 1s 2ms/step - accuracy: 0.9173 - loss: 0.4707
✅ Test Accuracy: 0.9208

 Test Loss: 0.4762

```
y_pred_prob = model.predict(X_test)

y_pred = (y_pred_prob > 0.5).astype(int)

for i in range(10):
```